Helsinki Smart City Mini-Platform

Business Case — Portfolio Project

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RMSE	MAE	R^2	Sensors	Forecast
6.97	5.30	0.788	3	12h
AQI units	AQI units	score	IoT nodes	LSTM ahead

About This Project

Air quality is something every city resident cares about, yet most people have no idea what the air is actually like on their street right now. Helsinki has nine fixed monitoring stations spread across the entire metropolitan area. That sounds reasonable until you realise it leaves the vast majority of streets, schools, and neighbourhoods without any measurement at all.

I built this platform to show what a modern, affordable alternative looks like. It simulates a network of low-cost IoT sensors across three Helsinki districts, streams their readings into a cloud database, and uses a machine learning model to predict air quality twelve hours into the future. Everything is visible in a live dashboard that anyone can open in a browser.

This document explains the thinking behind it, what it does technically, and where it could go next.

1. The Problem Worth Solving

Walk down Fleminginkatu in Kallio on a Tuesday morning during rush hour and the air quality is measurably worse than it was at 7am. But nobody knows that in real time, because the nearest HSY monitoring station is over two kilometres away and only reports once an hour. That gap between what we can measure and what we actually need to know is the core problem this project addresses.

There are three specific issues with how things work today:

Issue	Current situation	Why it matters
Coverage	9 stations across the whole city	About 85% of streets have no reading at all
Update speed	Readings every hour	Short pollution spikes from traffic are invisible
No forecasting	No predictive capability exists	Authorities can only react, never prepare
Expansion cost	Each new station costs €15,000 to €40,000	Growing the network is very slow

2. What I Built

The platform has four layers that work together. Sensors generate readings every thirty minutes and send them to a cloud database. A machine learning model reads those historical patterns and produces a twelve-hour forecast. A dashboard makes all of this visible and understandable to anyone, not just data scientists.

Layer	What it does	Technology
IoT sensors	Three district sensors stream air quality, traffic and energy data every 30 minutes	Python, MQTT
Cloud storage	All readings are stored and indexed so any time range can be queried instantly	MongoDB Atlas
LSTM model	Learns daily and weekly patterns then forecasts AQI twelve hours ahead with confidence bands	TensorFlow, scikit-learn
Live dashboard	Charts, heatmap by hour and district, threshold alerts, and the forecast overlay	Streamlit, Plotly

How accurate is the forecast?

The model was trained on sixty days of sensor data at thirty-minute intervals. It looks back twenty-four hours to predict the next twelve. The numbers below show it performs well enough to be genuinely useful for city-scale decisions.

Metric	Value	Plain English meaning

RMSE	6.97 AQI	Typical forecast error is below the threshold the WHO considers significant
MAE	5.30 AQI	On average the prediction is off by less than one AQI category
R^2 score	0.788	The model explains about 79% of the variation in air quality

3. Who Would Use This

Several groups would find immediate value in a platform like this. The table below summarises who they are and what they get from it.

Who	What they get
Helsinki city authority	Ready-made EU compliance reports, granular planning data, and automated public health alerts without expanding the physical station network
Regional health agencies	Early warning when air quality in a specific district crosses a safe threshold, with twelve hours to prepare a response
Urban planners	Hyper-local data at the street level to inform decisions about traffic routing, green space, and zoning
Real-estate developers	Environmental data to support ESG reporting and site due diligence
Residents	A free, always-on dashboard that shows what the air is actually like in their neighbourhood right now

4. Why This Approach Makes Sense

Traditional air quality stations are precision instruments built for regulatory compliance. They are accurate, certified, and expensive to install and maintain. That makes sense when you need legally defensible measurements at a small number of fixed reference points.

This platform is designed for a different purpose: dense, real-time spatial coverage across a neighbourhood rather than a single certified reading per district. The components it uses, open-source software, a free-tier cloud database, and commodity hardware for physical deployment, are all either free or very low cost. Nothing in this stack requires a procurement process or a specialised vendor.

The practical implication is that a city or research team could deploy this alongside the existing HSY network rather than instead of it. The certified stations continue doing what they do well. This platform fills the gaps between them, adds a forecasting layer, and makes the data accessible to non-specialists through the dashboard.

What makes it deployable

Aspect	How this platform handles it
Software cost	Entirely open source. Python, TensorFlow, MongoDB Community, and Streamlit are all free to use.
Cloud storage	MongoDB Atlas free tier is sufficient for a pilot. Paid tiers scale as data volume grows.
Hardware	The current version uses simulated sensors. Physical deployment would use low-cost IoT boards such as Raspberry Pi with air quality sensors.

Maintenance	No specialist infrastructure team needed. The dashboard and model run on standard cloud services.
Data access	The HSY open data API is free and publicly available. Connecting real Helsinki readings requires only a configuration change.

5. What Comes Next

The current version is a fully working portfolio prototype. The sensor data is simulated, the model is trained and deployed, and the dashboard is live. The natural next step is to connect it to real Helsinki open data from HSY and then deploy a small set of physical sensors in one district as a proof of concept.

Phase	When	What	Status
IoT simulation	Week 1 to 2	Sensor simulator and MongoDB schema built and tested	Done
LSTM model	Week 3 to 4	Model trained, RMSE 6.97, R2 0.788, saved and integrated	Done
Live dashboard	Week 5 to 6	Streamlit app deployed on Streamlit Cloud with real forecast	Done
Business case	Week 7	This document written and included in the portfolio	Done
GitHub and LinkedIn	Week 8	Public repository, short demo video, and LinkedIn write-up	In progress
Real data connection	Q3 2025	Replace simulated data with live Helsinki HSY open data API	Planned
Hardware pilot	Q4 2025	Three Raspberry Pi sensors physically deployed in Kallio	Planned
Municipality proposal	Q1 2026	Formal pitch to HSY and a Horizon Europe Green Deal application	Planned

Why this project matters to me personally

My research background is in IoT-based air quality monitoring, where I worked on LSTM and spatial-temporal models for environmental prediction. This project is the natural continuation of that work, rebuilt with modern cloud tools and packaged in a way that a city planner or a health official could actually use, not just a dataset that sits in a journal.

The MBI programme at Metropolia gave me the frameworks to think about this not just as an engineering problem but as a service design and digital strategy challenge. The business case, the cost analysis, and the roadmap above came directly from applying that coursework to something I genuinely care about.

Helsinki felt like the right place to build this. The city has real ambitions around smart city infrastructure, open data, and sustainability. I wanted to show that those ambitions do not have to wait for large budgets or long procurement cycles.